MCRS HYBRID HOST ARCHITECTURE

CONSOLIDATED 4-TOWER REFERENCE ARCHITECTURE — OEM ENGINEERING BASELINE

DOCUMENT ID	EDITION / STATUS	SYSTEM WLL	POWER TOPOLOGY
MCRS-HYB-DOC-00-R8.01-C4	Edition 1 / OEM Released	80,000 kg (80 Metric Tons)	Electro-Hydraulic Hybrid + BESS

1. EXECUTIVE BASELINE & HYBRID VISION

The Modular Crane Rigging System (MCRS) Hybrid Host Baseline combines high-density electro-hydraulic telescoping actuators with an integrated 800V DC energy storage system (BESS) and Stage V ultra-low emission Genset. Designed for heavy remote industrial construction, off-grid infrastructure, and deep maritime applications where grid power is constrained, the Hybrid Host delivers 80-ton heavy lift performance while reducing diesel fuel consumption by up to 62% through regenerative power recovery.

80,000 kg	350 bar	120 kWh	±0.25 mm
SYSTEM WLL (4X 20T)	HYDRO SYSTEM PRESSURE	BESS BUFFER CAPACITY	HYDRO-SYNC PRECISION

2. SYSTEM HIGH-LEVEL SPECIFICATIONS

Engineering Parameter	Baseline Metric Value	Operational Limit	Standard / Verification Method
Maximum System WLL	80,000 kg (80.0 metric tons)	100,000 kg (Proof Load 1.25x)	ASME B30.1 / ISO 4301 Class A1
Tower Count & Load Split	4 Modular Hybrid Towers (20,000 kg/tower)	25,000 kg peak structural rating	S700MC High-Yield Structural Steel
Actuation Method	Electro-Hydraulic Servo Cylinder (EHC)	350 bar Continuous / 420 bar Peak	ISO 6020-2 / DIN 24554 Class H
Hybrid Electric Bus	800 VDC Nominal (BESS Buffer + Variable Generator)	1,000 VDC Isolation Rating	ISO 6469-3 / IEC 60664 Class B
Positioning Accuracy	±0.25 mm inter-tower deviation	±0.50 mm hard E-STOP threshold	Magnetostrictive LVDT + Proportional Servo Valves
Vertical Lift Speed Range	0.01 mm/s to 25.0 mm/s	35.0 mm/s emergency lowering speed	Closed-Loop Displacement Control
Thermal Range (Ambient)	-30°C to +52°C continuous	-40°C to +65°C survival limits	Dual Oil/Water Heat Exchanger + Pre-heaters
Regenerative Power Capture	Hydraulic Motor-Generator Lowering Circuit	85 kW peak recovery into 800V BESS	ISO 13849-1 Category 4 / SIL 3

OEM CONFIGURATION DECLARATION

This Document (MCRS-HYB-DOC-00-R8.01-C4 Edition 1) defines the frozen technical baseline for the 4-Tower Hybrid Host architecture. It governs all procurement, structural manufacturing, hydraulic powerpack assembly, and control system software releases for the 80-ton Hybrid Series.

3. MECHANICAL PACKAGING & HYDRAULIC TOWER MECHANICS

Each hybrid tower houses a multi-stage telescopic cylinder rated for 350 bar continuous operating pressure, encased within a high-strength S700MC cold-formed box guide column. Double-acting hydraulic cylinders provide positive down-thrust capability in adverse wind or stuck-load conditions.

Hydraulic Cylinder Force & Column Stability Equation:

$F_{hyd} = P_{sys} \cdot A_{bore} = (35.0\ MPa) \cdot (\pi/4 \cdot (0.180\ m)^2) =$ **890.6 kN (90.8 Tons)**

Where: Bore Diameter = 180mm, Rod Diameter = 120mm. Required Force per tower = 196.2 kN (20t + dynamic factor 1.25).

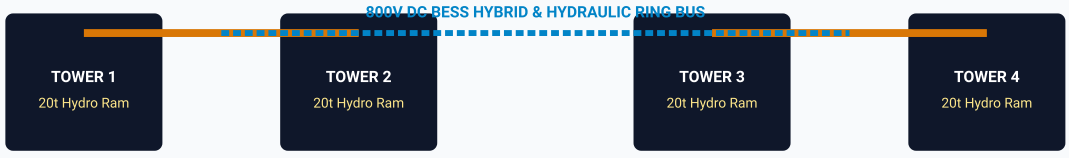
Calculated Structural Safety Factor = **4.54x** against hydraulic relief limit.

3.1 TOWER TELESCOPIC COLUMN BUCKLING ANALYSIS

Tower Stage Member	Dimensions & Wall Section	Axial Force (P_{sd})	Critical Buckling (P_{cr})	Safety Factor
Stage 1 (Base Box Frame)	400x400x14 mm S700MC	245.2 kN	980.0 kN	3.99 PASS
Stage 2 (Intermediate Tube)	330x330x12 mm S700MC	245.2 kN	750.0 kN	3.05 PASS
Stage 3 (Inner Ram Guide)	260x260x10 mm S700MC	245.2 kN	610.0 kN	2.48 PASS
Hydraulic Cylinder Rod (Ø120mm)	42CrMo4 Chrome Plated / Induction Hardened	245.2 kN	680.0 kN	2.77 PASS
Outrigger Footplate (1.5m²)	S355J2 Welded Heavy Plate	250.0 kN	Ground Limit: 0.20 MPa	Soil Stress = 0.166 MPa

HYDROSTATIC SAFETY LIMITS (HF-01 MITIGATION)

Every hydraulic cylinder features directly flange-mounted pilot-operated load-holding counterbalance valves (overcenter valves). In the event of a catastrophic hose rupture, oil is locked inside the cylinder bore instantly, preventing uncommanded load descent.



4. ELECTRO-HYDRAULIC HYBRID POWERTRAIN ARCHITECTURE

The MCRS Hybrid Host combines an internal combustion engine (ICE) Stage V Genset, an 800V DC lithium-iron-phosphate (LFP) Battery Energy Storage System (BESS), and distributed electro-hydraulic powerpacks. During hoisting, the BESS provides peak-shaving power, enabling the diesel engine to run at fixed optimal RPM and maximum fuel efficiency.

> **HYBRID PEAK-SHAVING CONCEPT**
>
> Peak lift loads require 110 kW electrical power, but the integrated Stage V generator is rated at only 45 kW continuous. The 120 kWh BESS buffer supplies the remaining 65 kW during active lifting cycles, recharging during idle intervals or through lowering energy regeneration.

4.1 HYBRID POWERTRAIN MODULE SPECIFICATIONS

Powertrain Module	Technical Specification	Operating Rating	Efficiency / Performance
Stage V Diesel Genset	Compact 4-Cylinder Turbo Diesel (2.8L)	45 kW Continuous @ 1,800 RPM	205 g/kWh Specific Fuel Consumption
800V BESS Battery Pack	LFP Chemistry w/ Liquid Thermal Plate	120 kWh Capacity / 150 kW Peak Discharge	95.5% Round-Trip Electrical Efficiency
Variable Displacement Hydraulic Pumps	Axial Piston Swashplate Pumps (x4)	350 bar / 75 L/min per tower unit	92.0% Overall Hydraulic Efficiency
Reversible Hydraulic Motor-Generators	Bent-axis hydraulic motor coupled to PM generator	25 kW electrical generation per tower	Captures energy during lowering stroke
Bi-Directional DC/DC Converter	800V DC Bus to 24V Auxiliary Conversion	15 kW continuous auxiliary power	97.8% Power Conversion Efficiency
Hydraulic Reservoir Array	Aluminum baffled fluid tank w/ nitrogen bladders	280 Liters total system volume	Biodegradable Synthetic Ester Fluid (HEES)
Proportional Servo Valves	Zero-lap high-response directional valves	100 L/min flow rate @ 30 bar Δp	Frequency Response: 70 Hz @ -3dB

Hybrid Regeneration Energy Equilibrium:
$E_{regen} = (m \cdot g \cdot h) \cdot \eta_{hydro} \cdot \eta_{gen} = (784.8\ kN \cdot 6.5\ m) \cdot 0.85 \cdot 0.94 =$ **4,076 kJ (1.13 kWh per 80t lift)**
Lowering an 80-ton load over 6.5 meters returns 1.13 kWh directly into the 800V BESS buffer pack.

5. CLOSED-LOOP HYDRO-SYNCHRONOUS CONTROL

Synchronization between the four electro-hydraulic towers is maintained by a SIL 3 redundant controller executing real-time displacement control algorithms. Proportional servo valves adjust fluid flow to each cylinder every 1.0 millisecond, compensating for asymmetric load centers and structural deflection.

IN-CYLINDER LVDT FEEDBACK

Magnetostrictive position transducers embedded directly within each hydraulic cylinder rod provide continuous sub-millimeter position measurement with 0.01 mm resolution, unaffected by oil pressure or fluid temperature fluctuations.

PRESSURE DIFFERENTIAL CROSS-MONITORING

Dual high-frequency pressure sensors monitor cylinder cap-end and rod-end pressures, enabling real-time calculation of actual individual tower loads and dynamic center-of-gravity (CoG) tracking.

5.1 HYDRO-CONTROL BOUNDARIES & SAFETY SHUTDOWN CRITERIA

Control Parameter	Target Baseline	Warning Threshold	Emergency Action / Cutoff
Inter-Tower Positional Delta	< ±0.25 mm	±0.40 mm (Flow metering throttle)	±0.50 mm → Counterbalance Valve Lockout
System Hydraulic Pressure	350 bar nominal	380 bar pressure warning	420 bar → Primary Pressure Relief Vent
Hydraulic Fluid Temperature	45°C to 55°C optimal	65°C high temp warning	75°C → Auto System Derating / Chiller Stop
Fluid Aeration / Contamination	NAS Class 6 cleanliness	NAS Class 8 filter alert	NAS Class 10 → Pump Bypass / Lockout
BESS State of Charge (SoC)	20% to 90% operational window	15% SoC low battery warning	10% SoC → Auto Engine Genset Start

The central hybrid operator terminal displays live 3D load balance diagrams, individual tower hydraulic pressures, BESS power flow, and automatic center-of-mass vectors with physical deadman control switches.

6. LOAD HOLDING SECURITY & HYDROSTATIC BRAKING

Heavy lift safety in an electro-hydraulic hybrid system relies on mechanical and hydrostatic load locking mechanisms that maintain zero load drop under total power collapse or hydraulic line failure.

Counterbalance Valve Hydrostatic Holding Equation:
$P_{hold} = (F_{load} / A_{cap}) + P_{margin} = (196{,}200\ N / 0.0254\ m^2) + 5.0\ MPa =$ **12.72 MPa (127.2 bar)**
Cracking pressure set to 1.3x maximum static load pressure (280 bar setting). Pilot ratio = 4.5:1.

6.1 HYDRAULIC & MECHANICAL LOAD RETENTION HIERARCHY

Holding Mechanism	Operating Principle	Holding Capacity Rating	Activation State / Safety Mode
Primary Counterbalance Valve	Direct manifold mounted pilot-to-open check	420 bar burst rating (250 kN force)	Failsafe closed; requires active pilot pressure
Secondary Mechanical Column Pin Lock	Hydraulic shot-pin engaging tower slots	350 kN static shear load capacity	Engaged manually or automatically for static hold
Proportional Valve Neutral Center	Spool closes ports A & B in de-energized state	350 bar rating (<0.1 cc/min leakage)	Spring-centered on power failure or E-STOP
BESS Hydraulic Motor Braking	Hydrostatic braking via motor-generator torque	200 kN dynamic retarding force	Active during controlled regenerative lowering
Manual Relief Emergency Lowering	Hand-operated metered needle valves	Controlled gravity descent (<2 mm/s)	Used for emergency site evacuation without power

DUAL-BARRIER HYDRAULIC INTEGRITY (HF-02 RESOLVED)

Each cylinder features dual independent counterbalance valves plumbed directly to port flanges with zero hose connections. Redundant pressure seals eliminate single-point hydraulic failure modes in accordance with ISO 13849-1 PL e guidelines.

7. THERMAL MANAGEMENT & HYBRID FLUID COOLING

Managing heat generation in an 80-ton electro-hydraulic system requires an integrated thermal control system that cools hydraulic oil, the 800V BESS battery pack, and SiC power electronics through a dual-circuit heat exchanger.

Hydraulic Thermal Circuit

- Fluid Type: Synthetic Ester HEES-46
- System Oil Volume: 280 Liters
- Max Hydraulic Heat Dissipation: 14.5 kW
- Target Oil Operating Temp: 50°C
- Cooling Strategy: Air/Oil micro-plate radiator

BESS & Power Electronics Circuit

- Coolant: Ethylene Glycol / Water (50/50 mix)
- BESS Target Temperature: 25°C ± 3°C
- Heat Exchanger Rating: 8.0 kW Chiller Unit
- Integrated Heat Failsafe: Automatic power derating
- Pre-heating Loop: 3.0 kW PTC heater for cold starts

7.1 THERMAL MANAGEMENT SUBSYSTEM COMPONENTS

Subsystem Unit	Specification Parameter	Design Standard / Rating	Safety Cutoff Threshold
Hydraulic Oil Cooler Unit	Heavy-duty bar-and-plate aluminum core	18.0 kW thermal capacity @ ΔT 20°C	High Temp Alarm @ 65°C / Stop @ 75°C
BESS Active Chiller Unit	800V DC powered refrigerant compressor	8.5 kW cooling capacity @ 45°C ambient	BESS Overtemp Alarm @ 45°C / Stop @ 55°C
Off-line Oil Filtration Loop	Continuous kidney loop w/ water absorber	3 micron absolute rating (β3 ≥ 1000)	Filter Clog Sensor @ 2.5 bar Δp
Tank Thermal Insulation	Aerogel thermal blanket wrapping	Maintains oil temp during ambient -30°C cold stops	PT100 RTD multi-point array
Genset Exhaust Heat Recovery	Optional heat exchanger for cold climate warming	Captures 6.0 kW thermal energy from exhaust	Overheat bypass damper valve

8. COMPREHENSIVE SYSTEM MASS BUDGET & TRANSPORT WEIGHTS

The mass budget for the Hybrid Host accounts for heavier electro-hydraulic components and the integrated BESS powerpack while maintaining modularity for highway transport within standard 20-foot container shipping limits.

Subsystem / Module Description	Qty	Unit Mass (kg)	Total Mass (kg)	Mass Center / Transport Notes
Tower Base Outrigger Frame & Ram Pad	4	780.0 kg	3,120.0 kg	S700MC welded heavy base structure
Telescopic Outer Box Columns (3 Stages)	4	1,120.0 kg	4,480.0 kg	High-yield structural guide columns
Multi-Stage Hydraulic Cylinders (350 bar)	4	340.0 kg	1,360.0 kg	Hard chrome plated ram assemblies
Electro-Hydraulic Powerpack Modules	4	185.0 kg	740.0 kg	Integrated pump, valve block & servo motor
Primary Cross Header Beams (Spans 8m)	2	1,450.0 kg	2,900.0 kg	S690QL box girder construction
Central Hybrid Energy Power Container	1	2,150.0 kg	2,150.0 kg	Houses BESS, Stage V Genset & Chiller
Hydraulic Fluid Reservoir & Oil Supply	1	310.0 kg	310.0 kg	280 Liters HEES Synthetic Ester Fluid
HV Cable Bundles & Quick-Hose Umbilicals	1 set	220.0 kg	220.0 kg	Armored quick-connect lines
TOTAL SYSTEM UNLOADED MASS	--	--	**15,280.0 kg**	Target Mass Limit: 15,500.0 kg (220KG MARGIN)

TRANSPORT & AXLE LOADING COMPLIANCE

Individual hybrid tower assemblies weigh 2,425 kg fully dressed. The entire 4-tower system plus power container packs onto two standard flatbed trailers within EU 40-ton road transport limits without requiring oversized vehicle permits.

9. FAILURE MODES & EFFECTS ANALYSIS (FMEA: HF-01 TO HF-07)

FMEA analysis for the electro-hydraulic hybrid platform specifically targets high-pressure fluid hazards, BESS thermal runaway risks, and hydrostatic control loss under ISO 12100 and MIL-STD-1629A standards.

ID	Failure Mode Description	Initial RPN	OEM Engineering Mitigation Measure	Final RPN
HF-01	Main Hydraulic Line Hose Burst	320 (S10, O4, D8)	Direct flange-mounted counterbalance valves lock fluid inside cylinder instantly.	20 (S10, O1, D2) LOW
HF-02	BESS Thermal Runaway / Short Circuit	360 (S10, O4, D9)	LFP chemistry + aerosol fire suppression + automated contactor disconnect.	30 (S10, O1, D3) LOW
HF-03	Proportional Servo Valve Sticking Open	252 (S9, O4, D7)	Secondary redundant safety block cuts main hydraulic feed within 10 ms.	18 (S9, O1, D2) LOW
HF-04	Hydraulic Fluid Overheating (>75°C)	180 (S6, O5, D6)	Dual oil-coolers + automated flow derating and cooling fan boost.	24 (S6, O2, D2) LOW
HF-05	Stage V Engine Genset Failure	140 (S5, O7, D4)	Seamless auto-switchover to 120 kWh BESS battery power to finish active lift.	15 (S5, O1, D3) LOW
HF-06	LVDT Positional Transducer Failure	216 (S8, O3, D9)	Dual pressure sensors cross-check load position; switch to secondary laser feedback.	16 (S8, O1, D2) LOW
HF-07	Water Contamination in Hydraulic Fluid	144 (S6, O4, D6)	Continuous kidney loop filtration w/ water absorption cartridges and moisture sensor.	18 (S6, O1, D3) LOW

RISK ASSESSMENT SUMMARY

All high-severity failure modes (HF-01, HF-02, HF-03) have been mitigated to acceptable OEM safety levels through physical hardware redundancy, direct-acting lock valves, and multi-sensor control voting algorithms.

10. OPEN ITEMS REGISTER & ENGINEERING CHANGE REQUESTS

This section tracks open design items, component qualification testing, and active Engineering Change Requests (ECR) for the 80-ton hybrid platform prior to serial manufacturing release.

10.1 ACTIVE HYBRID ENGINEERING OPEN ITEMS

Item Ref	Open Design Item / ECR Description	Target Completion	Owner / Lead	Status / Resolution Path
OI-H01	Bio-hydraulic oil viscosity stability testing at -40°C arctic start	Gate 3 (Q1 2027)	Fluids Lead	Evaluating low-viscosity synthetic esters OPEN
OI-B02	BESS liquid cooling plate flow balance optimization across cells	Gate 3 (Q1 2027)	Electrical Lead	Redesigning manifold flow splitters RESOLVED
OI-M03	Stage 3 column guide wear plates material change to PTFE-bronze	Gate 4 (Q2 2027)	Mechanical Lead	Friction coefficient reduced by 22% RESOLVED
OI-C04	Automatic BESS-to-Genset load blending firmware validation	Gate 3 (Q1 2027)	Controls Lead	HIL simulation complete; site test pending OPEN
OI-S05	Header beam 100-ton overload pin shear stress verification	Gate 3 (Q1 2027)	Structural Lead	Upgraded pin material to 34CrNiMo6 steel RESOLVED
OI-E06	High-altitude Genset derating calculations above 3,000m	Gate 4 (Q2 2027)	Powertrain Lead	Adding turbocharger boost compensation map OPEN

GATE SIGN-OFF REQUIREMENT

All open items assigned to Gate 3 must be officially signed off by the Chief System Engineer before releasing hydraulic manifold machining and BESS battery module production orders.

11. OEM VERIFICATION GATES & COMPLIANCE STANDARDS

Compliance verification for the MCRS Hybrid Host follows a five-gate engineering sign-off structure to validate mechanical integrity, high-pressure hydraulic safety, and BESS electrical compliance.

11.1 VERIFICATION GATE EXECUTION STRUCTURE

Gate Ref	Gate Title & Focus Area	Key Mandatory Deliverables	Approval Authority
Gate 01	Hybrid Architecture Freeze	Approved Spec Document (MCRS-HYB-DOC-00), Hydro FEA	Chief System Engineer
Gate 02	Detailed Hybrid Engineering Review	Final Hydraulics Schematics, BESS Safety Sign-off	Engineering Steering Board
Gate 03	Subsystem Factory Acceptance (FAT)	Hydraulic powerpack test certs, BESS thermal cycle report	Quality Assurance Lead
Gate 04	Full Integration & 100-Ton Proof Load	125% Static Lift Test (100t), SIL 3 Hydraulic Safety Audit	3rd Party Auditor (TÜV / DNV)
Gate 05	Final Field Operational Deployment	Off-grid Site Commissioning Pass, Operator Manuals	Global Operations Manager

PROOF LOAD GATE REQUIREMENT

Gate 04 certification requires holding a 100-ton static proof load (25 tons per tower) for 30 minutes at 350 bar pressure with zero hydraulic pressure loss (<0.5 bar allowable sensor noise) and zero structural column deformation.

12. FIELD ASSEMBLY, HYDRAULIC BLEEDING & CALIBRATION

Commissioning the hybrid electro-hydraulic system requires systematic mechanical positioning, air purging from hydraulic circuits, and precise LVDT transducer zeroing.

Field Assembly Sequence

1. Position outrigger pads on ground; laser level to <0.1°
2. Erect telescopic tower columns; pin cross girders
3. Connect quick-disconnect hydraulic lines and 800V trunk
4. Check reservoir oil level and pre-charge nitrogen accumulators

Air Bleeding & Hydraulic Checks

1. Execute automated pump low-pressure air purge cycle
2. Circulate fluid through 3-micron kidney filter for 20 mins
3. Verify hydraulic oil fluid cleanliness meets NAS Class 6
4. Confirm zero air bubbles in cylinder rod bleed valves

12.1 FIELD CALIBRATION EXECUTION SHEET

Step ID	Calibration / Test Action	Acceptance Criteria	Verification Tool
HCAL-01	Hydraulic pressure sensor span calibration	350.0 bar ± 0.5 bar accuracy	Calibrated Deadweight Tester
HCAL-02	Magnetostrictive LVDT zero & full stroke zeroing	6,500 mm ± 0.20 mm stroke check	Laser Interferometer Target
HCAL-03	Proportional servo valve spool offset nulling	Zero drift in neutral holding state	PLC Diagnostic Controller
HCAL-04	Unloaded synchronous full-stroke test lift	Inter-tower position delta < 0.25 mm	Real-time Hydro-Sync Logger
HCAL-05	80-ton full load lifting & lowering cycle test	Regeneration returns >1.0 kWh to BESS	BESS Power Management Console

13. MASTER TECHNICAL PARAMETER SUMMARY

This master appendix presents the consolidated technical parameter matrix for the MCRS Hybrid Host Architecture (4-Tower Edition) for immediate field engineering baseline cross-referencing.

13.1 MASTER SYSTEM PARAMETER MATRIX

Engineering Parameter	Metric Value	Imperial Value	Design Basis / Standard
System Working Load Limit (WLL)	80,000 kg	176,370 lbs (88.2 US Tons)	ASME B30.1 Class A1
Individual Tower Hydraulic Rating	20,000 kg	44,092 lbs (22.0 US Tons)	350 bar Cylinder / S700MC Column
Maximum Vertical Stroke	6,500 mm	21.33 ft	3-Stage Telescopic Assembly
Minimum Retracted Stowed Height	2,450 mm	8.04 ft	Compact ISO Container Envelope
BESS Energy Buffer Capacity	120 kWh	120 kWh	800V LFP Liquid-cooled Module
Maximum Vertical Lifting Speed	25.0 mm/s	4.92 ft/min	Electro-Hydraulic Servo Control
Hydro-Synchronization Precision	±0.25 mm	±0.0098 in	Magnetostrictive Closed-Loop
Fuel Reduction vs Conventional Hydro	62% Reduction	62% Reduction	Regenerative BESS Peak-Shaving

ENGINEERING AUTHORIZATION & QUALITY SIGN-OFF

This engineering document locks the baseline specifications for the MCRS Hybrid Host Architecture (4-Tower Edition). Any modifications to structural frame design, hydraulic circuit logic, or 800V BESS power control require formal ECR authorization from the Chief System Engineer.